

Day 02 - Linux Permissions, Users & SSH

Objectives

Understand Linux permissions, ownership, users, groups, sudo and SSH authentication. Learn to troubleshoot Permission Denied and SSH login failures.

Memory Tricks

chmod = CHange MODe (changes permissions)

chown = CHange OWNeR (changes ownership)

WHO owns the file? → **chown**

WHAT can the file do? → **chmod**

Permission values: r=4, w=2, x=1

rwX=7, rw-=6, r-x=5, r--=4, ---=0

Important Commands

whoami – current user

id – user, UID, GID and groups

groups – list groups

ls -l – view permissions

chmod +x deploy.sh

chmod 755 deploy.sh

chown ubuntu:ubuntu deploy.sh

ssh -i key.pem ubuntu@PUBLIC_IP

Interview Questions

- What does chmod 755 mean?
- Difference between chmod and chown?
- Difference between whoami, id and groups?
- Why use SSH keys instead of passwords?
- Why use sudo instead of logging in as root?

Production Scenarios

Permission denied: check ls -l, owner, permissions, then fix.

Permission denied (publickey): verify username → key → Security Group → SSH service → authorized_keys.

Best Practices

Investigate first. Fix second.

Follow least privilege.

Never share PEM files.

One SSH key per developer.

Key Takeaways

chmod changes permissions.

chown changes ownership.

Use groups to manage access.

SSH keys are more secure than passwords.

Investigate → Understand → Fix.